

Edouard Levasseur

📍 Sheffield | ✉ ed.t.levasseur@gmail.com | 📞 07384 021 628 | 🌐 lewardo.dev | 🌐 in/lewardo | 🐦 @lewardo

Personal Profile

Second year university student studying BSc. Computer Science at the *University of Sheffield*; working part time as an Embedded C++ and CMake Build System Developer at *Ossila*, and intern at the ERC Horizon-funded *FluCoMa* research project.

A background in professional development with a focus on embedded systems and machine learning, and experience working as part of a team and independently in a professional environment. Working on many personal projects developing my knowledge, motivation and experience.

Education

University of Sheffield | BSc. Computer Science Sept 2023 – Present

- In studies, graduating Jun 2026 with projected 1st class degree
- *Specialisation*: Machine Vision, Algorithmics, Machine Learning
- *Dissertation*: *Investigating Embedded DSP Pipelines for Standalone Stroke Reconstruction*; involving Embedded Systems, Machine Learning, Sensor Fusion, DSP

Greenhead College | A-Level Qualifications Sep. 2021 – Jun. 2023

- A-Levels: Maths (A*), Further Maths (A*), Physics (A*), Philosophy (A*)
- Extended Project Qualification: *On the Low-Cost Production of Concealed Augmented Reality Systems* (A*, full marks)

Experience

In chronological order

Embedded C++ and CMake Build System Engineer | Ossila Ltd., Sheffield Apr. 2024 –

- Hardware abstraction and build system engineer
- Built a complete build system in CMake from the ground up that manages hardware abstractions, libraries, linting and formatting, updates, and more
- Currently working on writing a bare-metal hardware abstraction framework for various chipsets (SAM3X, SAMD51, RP2350, STM32), involving writing low-level hardware drivers and bare-metal abstraction
- *Technologies used*: C++, CMake, git, gcc, clang, LinkerScript, Python3, Atmel SAMD51, RPi RP2040/2350, STM32

Parkour Club Captain | University of Sheffield Jun. 2024 –

- Captain of the University of Sheffield Parkour Club, with wide ranging responsibilities from management to outreach
- Ran stalls at sports fairs, taught parkour at many experience levels, organised social gatherings, designed and distributed merchandise, managed sponsorships and relations with the wider Club Sport body, and more
- Organised outreach initiatives, open sessions, and inclusive efforts to grow the club

C++ ML Developer | FluCoMa Research Project, Huddersfield Aug. – Sep. 202{3,4}

- Interned at the Fluid Corpus Manipulation (FluCoMa) Horizon Research project (grant agreement No 725899) implementing various machine learning algorithms in C++, including dynamic time warping, recurrent neural network architectures, regression algorithms, and more
- Also maintained and refactored code for various other smaller projects in the department
- Experience with production-grade C++, optimising and implementing machine learning algorithms, UI/UX designing for musicians, using CMake, and Cycling 74's Max/MSP framework
- Worked both in a team and independently in a professional environment, and the flexible schedule developed my time management and organisation
- *Technologies used*: C++17, CMake, git, Max/MSP, SuperCollider, PureData

Avionics and Control Systems Engineer | Student Drone Projects (Various), Sheffield Oct. 2023 – Jun. 2024

- Part of the Missions Avionics subteam in Project Falcon and the Guidance, Navigation and Control subteam in Project Kestrel
- Developed the long-range wireless telemetry systems for the drone, designing and implementing packet generation, serialisation, and optimising reliability and throughput
- Both projects worked to develop fully autonomous fixed-wing aircraft, to compete in the IMechE UAS Challenge 2024

- Experience involving iterative testing, teamwork, communicating with other subteams for requirements
- *Technologies used:* C++, git, PlatformIO, ArduPilot

Freelance Frontend Developer | Electric Spring Festival, Huddersfield

Dec. 2022, Nov. 2023

- Commissioned to create a responsive website for the festival from a graphic concept in 2023 and 2024
- Experience working in a professional web development context with a client brief and requests, improved my communication, planning, and management skills
- *Technologies used:* HTML5/CSS3, JavaScript, JQuery, git

Freelance Frontend Developer | Sexpression:NCL, Newcastle

Aug. 2023 – Nov. 2023

- Commissioned by the Newcastle regional branch of Sexpression:UK to create their annual National Conference's website from a graphic concept
- Resulted in the charity National Director offering me the National Technical Officer role
- *Technologies used:* HTML5/CSS3/ES6, Vue3, Vite, WebPack, npm, git

Projects

Concealed Augmented Reality Smart Glasses

 lewardo/spectacle_fw

- Research project researching ways to create affordable, accessible, concealed augmented reality smart glasses
- Received funding from the Emerge: Future Founders Fund and shortlisted for the Emerge: Awards Showcase
- Experience programming STM32s and ESP32s, using FreeRTOS, low-level and wireless programming
- Mixed signal schematic and PCB design, and SMD assembly and hardware debugging
- *Technologies used:* C, FreeRTOS, git, Arduino, esp-idf, EasyEDA, ESP32, stm32

Vue3 Glitching Component

 lewardo/vue-glitched

- Simple glitching component for Vue 3 applications to apply a visual effect
- Fully configurable in every aspect, including responsive content and attribute-based
- NPM package has amassed over 2500 downloads since first release
- *Technologies used:* Vue3, Vite, JavaScript, git, npm

C++ Machine Learning Library (ongoing)

 lewardo/libml

- A simple machine learning library implemented from the ground up as an exercise to learn the new features of C++20
- Templated BLAS linalg implementation, expression templates,
- *Technologies used:* C++, CMake, git

Portfolio Website

 lewardo/lewardodotdev

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- *Technologies used:* Vue3, Vite, JavaScript, git, npm

Technologies

- *Languages:* C, C++, CMake, Java, Python, HTML/CSS, JavaScript, Bash; with some experience in Dart, Ruby, NodeJS
- *Frameworks:* Arduino, FreeRTOS, Espressif esp-idf, JQuery, Vue3
- *Specialisations:* Embedded C/C++ Systems, Machine Learning
- *Tools:* git, bash/Linux, vi, VSCode, Visual Studio, XCode

Qualifications and Skills

- *First Aid:* Royal Life Saving Society Emergency First Aid at Work (RLSS EFAW) and Automatic External Defibrillator (RLSS AED) certified
- *Languages:* Native French-English bilingual, both being spoken to first language fluency
- *Scouting:* Chief Scout Diamond Award and Duke of Edinburgh Silver Award
- *Olympiads:* Distinction in the UKMT BMO 2023 Round 1